

BSc (Hons) in Information Technology Specialized in AI
IT3012 : Intelligent Agents
 Year 3 - Semester 1 - 2026 Semester 2

Objective & Lecture Mapping: Build a goal-based planning agent using Breadth-First Search (BFS), Depth-First Search (DFS), and Uniform-Cost Search (UCS).

Part 1: Practical Implementation

- Implemented BFS with a FIFO queue and DFS with a LIFO stack.
- Implemented UCS with a priority queue and accumulated path cost.
- Added graph-search reached sets and exposed grid size, walls and food through the environment percept.

Part 2: Analytical Evaluation - Completed Answers

1. What is the difference between a state space and a search tree?

Answer: The state space contains the possible world states and legal transitions. A search tree contains the paths generated by an algorithm while exploring that space; the same state can occur at multiple tree nodes through different paths.

2. Why does DFS need a reached set?

Answer: The reached set prevents repeated expansion and cycles. Without it, DFS can keep revisiting the same grid cells and may never reach the goal.

3. Why is BFS optimal here while DFS is not?

Answer: Each grid move has unit cost. BFS explores paths in increasing depth, so the first solution uses the fewest moves. DFS follows one deep branch and may return a much longer solution before examining a shorter branch.

4. Compare BFS and DFS memory on a 1000 x 1000 grid.

Answer: BFS can hold a wide frontier and requires roughly $O(b^d)$ space. DFS mainly retains the current path and unexplored siblings, roughly $O(bm)$, so it normally uses less memory but may follow a very deep unhelpful path.

Completion and Verification

The implementation is stored in the matching Week_03 branch. Automated tests pass and the runnable visual environment remains available in visual_grid_game.py.